

Construction of enterprise accounting information system based on REA model

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ABSTRACT

The reconstruction of traditional enterprise accounting information systems has become a general trend in an environment where accounting policies are gradually opening up and information technology is developing rapidly. This study first explains the limitations of the traditional accounting system and the construction principles of the enterprise accounting information system based on the expansion of the REA model. On this basis, facing the actual sales collection business, it comprehensively analyzes the reconstruction process of the enterprise's multi-dimensional double-entry accounting information system to solve the current accounting problem. The problems existing in the information system are expected to provide reference for the promotion and application of the REA accounting model in the digital information era, as well as the research on enterprise information integrated management.

CCS CONCEPTS

• **Information systems** → Information systems applications; Enterprise information systems; Enterprise applications.

KEYWORDS

Multidimensional accounting model, Accounting information system, Data warehouse

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1 INTRODUCTION

With the rapid rise of new generation information technologies such as big data and artificial intelligence, China has entered the digital economy era. Both the management methods and operational models of enterprises are undergoing rapid changes, and the users of enterprise accounting information are increasingly showing personalized and diversified demands for accounting information. Although traditional accounting information systems have improved the efficiency of manual accounting work, they are

unable to address issues such as limited information coverage and information distortion, nor can they meet the information support needs for enterprise management and decision-making in the new information environment. The REA model describes the common economic phenomena involved in enterprise business processes through major entities such as resources, events, and agents, providing a more intuitive, comprehensive, and accurate reflection of financial and non-financial information in business events [1, 2]. As a new type of multidimensional double-entry accounting information system designed based on this model, it can provide richer information choices for managers at all levels of enterprises, enabling data users to conduct comprehensive and three-dimensional analysis and decision-making, demonstrating the potential characteristics of becoming the mainstream form of future applications for enterprise information systems and accounting systems [3]. Therefore, this study aims to reconstruct accounting information systems based on the extended application of the REA model to better utilize technology for improving management.

2 CONSTRUCTION APPROACH OF ENTERPRISE ACCOUNTING INFORMATION SYSTEM BASED ON REA

The basic idea of the REA model is to describe the economic transactions of an enterprise in its primitive form. Based on the E-R model, the REA model classifies the entities involved in enterprise economic activities into three categories: agents, events, and resources. Resources refer to quantifiable and scarce enterprise assets that can bring economic benefits [4, 5]. Events are business activities that cause changes in resources, such as purchases, sales, and production. Agents refer to departments, units, or individuals directly involved in economic events or influencing the activities of other participants. Supported by the theoretical foundation mentioned above, the construction of an accounting information system based on the REA model has a structure as shown in Figure 1.

As can be seen from Figure 1, the core idea and outstanding feature of the accounting information system based on REA is that the storage and use of data are completely determined based on the origin and process of economic business matters. When business events occur, the system uses modern information technology to collect three The original data of class entities is stored in the business database in real time. Information users submit event data applications and select information extraction rules. The system drives corresponding reporting tools to process the original data and generates them in a view format that meets the user's requirements. Information analysis report. In short, the accounting information system has realized the leap from department system to enterprise system under the construction of REA model, fully integrating operation management and accounting, information processing

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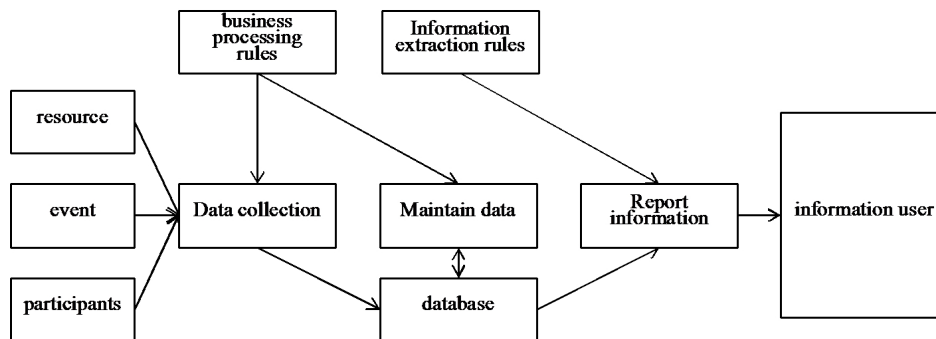


Figure 1: REA accounting system structure.

and business execution, non-financial information and financial information.

3 ANALYSIS OF CONSTRUCTION OF ENTERPRISE ACCOUNTING INFORMATION SYSTEM BASED ON REA

3.1 Requirement Analysis Stage

From the perspective of the enterprise, the sales receipt process is first divided, which is also the prerequisite for accurately analyzing information needs. The main stages of this process include: marketing, order acquisition, order verification, product stocking, outbound, transportation, order delivery, receipt, product return, or failed receipt. In investigating a series of stages, various personnel in enterprises have different information needs for the sales receipt process. Senior management mainly focuses on comprehensive sales information such as changes in bad debts, sales revenue, discounts and returns, changes in sales trends, and product market share to evaluate the effectiveness of business activities. Sales personnel are the main users of sales-related information, and their information needs include: sales amount, quantity and price, personal sales performance, regional sales differences, competitor sales information, customer type and preference analysis, etc.^[6]. Production personnel need to formulate and adjust product performance and production plans based on customer feedback information and customer order information. Finance and accounting personnel need long and short-term operational data such as inventory backlog, cash flow, and operating income to prepare financial reports and make investment decisions. Human resources personnel need to understand employee information in the sales and receipt process to evaluate employee performance and compensation, formulate personnel demand plans, etc.^[7]. In addition to meeting the needs of enterprise personnel, the information provided by the accounting information system should also meet basic system security requirements, such as reliability, integrity, confidentiality, etc., to prevent security risks such as data loss, damage, and theft.

3.2 Conceptual Design Stage

Firstly, identify important business events and clarify key issues such as event triggers. The identification of sales receipt process events is shown in Table 1.

Secondly, describe the clear entity relationships using the REA model diagram. The specific method is to determine the maximum and minimum values of relationship cardinality concepts between entities, and then reflect them in the REA model diagram. Taking the entity relationship between Product and Order Acceptance as an example, firstly determine the cardinality set of the Product. Since an order may involve multiple or only one type of product, the minimum cardinality of the Product entity is 1, and the maximum cardinality is n , thus the cardinality set of the Product entity can be represented as $(1,*)$. Then consider the cardinality set of the Order Acceptance event entity. A product may be ordered multiple times or not ordered at all, so the minimum cardinality of the Order Acceptance event related to the Product entity is 0, and the maximum cardinality is n , the cardinality set of an Order Acceptance event can be represented as $(0,*)$. From the above, it can be inferred that the relationship between Product and Order Acceptance entities can ultimately be represented as Product $(1,*)$ - $(0,*)$ Order Acceptance. Based on this example, the relationships between other types of entities can be further deduced, and then described in the REA model diagram, as shown in Figure 2.

3.3 Model Validity Verification

Opinions from personnel involved in various stages of the business process are solicited regarding the initial REA model diagram, to either modify the model or confirm its validity.

3.4 Logical Design Stage

A relational model is constructed to comprehensively reflect the logical structure of all relationships in the REA model, converting the REA conceptual model into a machine model. Specifically, based on the REA model diagram, the various business events and their related relationships in the sales receipt process are determined, along with corresponding primary keys, foreign keys, and table names. The specific details are presented in Table 2.

3.5 Physical Design Stage

Suitable hardware and software are utilized to implement the physical storage of data and enable convenient physical maintenance and retrieval operations such as creating indexes. This involves the practical application of the logical design of the REA-based system. Using the sales receipt cycle business as an example for analysis:

Table 1: Important business events.

event name	risk	trigger	Target
Promotional marketing	Unfavorable publicity, unreasonable marketing goals, etc.	business decisions	Stimulate customers to buy
accept order	Insufficient supply of orders, customers or sales personnel exceeding their authority to sign contracts, etc.	Customer decides	Products can meet order requirements
Order review	Customer's bad credit, etc.	Buy	Eliminate order credit risk
Product shipment	Warehouse personnel ship goods in violation of regulations, wrong products are shipped out of the warehouse, products are shipped out of the warehouse beyond their authority, and delivery locations are wrong, etc.	accept order	Reduce the delivery of products that meet order requirements
order delivery	The invoice is incorrect, the invoice is forgotten, the goods are lost in transportation, the delivery person or delivery address of the goods is wrong, the delivery is not timely, the shipper or the shipment event is not correct.	Order review	Ship, deliver and invoice products
Collect payment	Authorized, etc.	Product shipment	Collect money promptly
Product returns	Incorrect payment information records, misappropriation of cash, duplicate payments, etc.	Official delivery	Establish after-sales service credibility
Bad debt treatment	Illegal returns, products returned without storage, etc.	product	Reasonably evaluate sales performance

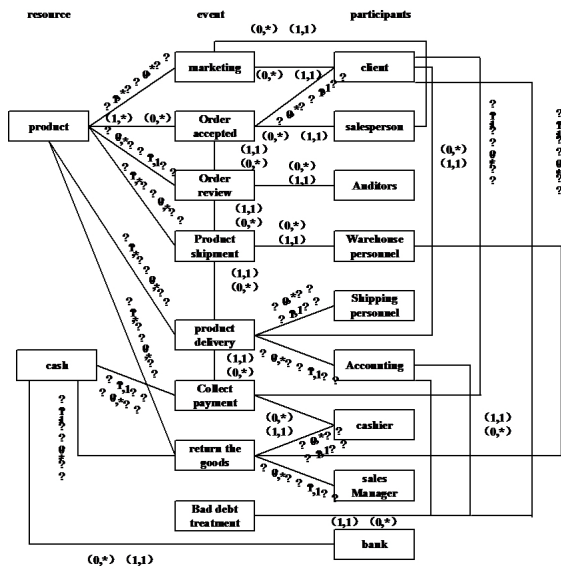


Figure 2: REA conceptual model diagram.

Based on the queried output tables of the system data, the involved entities include: participants (banks, customers, employees), resources (cash, products), and events (cash, sales). Sales-related records indicate that in Sales Event 002, Salesperson 009, Mr. B, on August 30, 2018, sold Product A (301) and Product C (302) to Customer H Company (203), resulting in a cash receipt of 20,000 yuan, marked as Cash Receipt 002. Based on the analysis of the data in Tables 1 to 10, it is evident that in Sales Event 002, Products A

and C generated a total income of 20,800 yuan, with a sales receipt of 20,000 yuan, leaving an outstanding amount of 800 yuan.

In conclusion, through the construction process of the REA accounting information system in the sales receipt business, it is apparent that the traditional accounting debit-credit relationships are intuitively reflected through the relationships between various data tables in the REA accounting system, providing a feasible approach to directly obtain accounting information from business details. The REA accounting system stores raw data directly in the database and describes and reflects the data through business event table views, greatly reducing the number of physical documents generated and processing steps involved in traditional accounting systems such as reports and postings, achieving convenient and simplified data usage effects, thereby optimizing timeliness and information quality.

4 CONCLUSION

Reconstructing a multidimensional double-entry accounting system not only aids in the semantic unity of information systems and accounting but also facilitates the integration of information systems with internal accounting within the enterprise. To ensure the new multidimensional double-entry accounting system model functions effectively, necessary safeguards should be provided in the following aspects.

Firstly, enhance system compatibility. During the construction of the new accounting information system model, thorough consideration must be given to software compatibility. It should be assessed whether inherent technologies can be compatible with the new system after the establishment of the multidimensional double-entry accounting system, ensuring that the application effectiveness of the new system is not affected by compatibility issues.

Table 2: Relational model table.

Logical Structure		Business Documents
Relation	Attributes	Correspondence Table
Marketing	[Marketing Event Number] (Salesperson Number) (Customer Number)	Product Marketing Sheet
Marketing/Product	[Product Number] [Marketing Event Number]	Marketing Schedule
Accept Order	(Salesperson Number) (Customer Number) [Order Number]	Order Form
Order/Product	[Product Number], [Order Number]	Order Details
Order Review	(Reviewer No.) (Order No.) [Review Event No.]	Audit Form
Review/Product	Approved Unit Price, [Product Number], [Audit Event Number]	Audit Detailed List
Product Shipment	(Audit Event Number) (Warehouse Personnel Number) [Warehouse Order Number]	Product Outbound List
Outbound/Product	Shipping Date, [Product Number], [Shipping Order Number]	Outbound Detailed List
Order Delivery	(Customer Number) (Account Number) (Outbound Order Number) (Shipper Number) [Invoice Number], [Shipping Event Number]	Product Delivery Form
Shipment/Product	[Product Number] [Shipping Event Number]	Product Delivery Schedule
Collect Payment	(Customer Number) (Cashier Number) (Account Number) [Cash Receipt Number]	Cash Table
Shipping/Collecting Payment	[Invoice Number] [Cash Receipt Number]	Income Statement
Return the Goods	(Cashier Number) (Account Number) (Sales Manager Number) (Account Number) [Return Event Number]	Return Form
Returns/Products	Return Quantity, [Product Number], [Product Number]	Return Details
Bad Debt Treatment	[Bad Debt Event Number] (Customer Number) (Invoice Number) (Employee Number)	Bad Debts
Participant Logical Structure		
Employee	Department, Position, Name, [Employee Number]	Employee Table
Client	Credit Limit, [Customer Number], Telephone	Customer Table
Bank	Contact Person, Address, Name, [Bank Number]	Bank Table
Resource Logical Structure		
Product	Quantity, Unit Price, Name, [Product Number]	Product List
Cash	Balance, Account Type, [Cash Account Number], (Bank Number)	Cash Table

Note: Square brackets are primary keys, and parentheses are foreign keys.

The reconstructed multidimensional double-entry accounting system should not only be fully compatible with the existing functions of the original accounting system but also meet the multi-level information needs of various users. Otherwise, the reconstructed system will encounter a series of problems during implementation. Firstly, strengthen communication and inclusiveness. Based on the industry characteristics and operational management status of the enterprise, select software systems and functional modules, fully considering whether the software systems and modules can meet future needs for functional integration, updates, and upgrades under the continuous development of the enterprise. Strictly compare similar software products from different service providers, as well as software products with differences in functionality, industry applicability, and understanding, and make choices carefully. Secondly, unify planning for accounting system software. Government departments, industry associations, or enterprise organizations should establish consistent agreements for accounting system construction. Relevant organizational departments should

coordinate various enterprise situations and provide accounting system software that adapts to the majority of enterprise situations. All enterprises should introduce and use them uniformly, enabling various enterprises to have a common accounting information processing method and form coherent accounting habits. Thirdly, refine parameter settings and strengthen development compatibility. Set reasonable parameters so that accounting personnel can ensure that the data input by other personnel complies with the system format requirements based on the accounting information system and establish corresponding restriction measures. Software developers should continuously research the compatibility of accounting system software, promote mutual compatibility between multidimensional accounting information systems and other internal management software within the enterprise through technological innovation, and for eligible enterprises, even collaborate with suitable software development companies to develop accounting system software suitable for the enterprise, considering the enterprise's specific situation.

Secondly, enhance the security of the system. As the supervisory department of enterprise accounting information, the finance department needs to take measures to ensure the security of its own department's systems. This includes establishing internal accounting system management regulations, clearly defining the roles and responsibilities of operators, and standardizing the behavior and permissions of operators. Establishing a system of operation records to prevent accounting data from being input or registered without verification, and incorporating accounting information system security management into the daily work guidelines of accounting personnel and integrating it into the enterprise management system to prevent accounting information from being stolen by external individuals through intrusion into the finance department network. If necessary, independent segmentation with other departments and external networks should be conducted. The maintenance of accounting system software and hardware should be entrusted to dedicated personnel to ensure the normal operation of the software and hardware devices that form the basis of the accounting system operation, which is the fundamental condition for the secure operation of the entire accounting system. To address the risk of information leakage, security passwords may be set for the core part of the accounting information system—the information database. Additionally, routine security maintenance of the accounting system is equally important. This includes regular cleaning and maintenance of system hardware devices, periodic testing of control systems and computer configurations, and continually strengthening the system's fault diagnosis capabilities and awareness of warning signs. Establishing an emergency response mechanism to ensure the safe recovery of enterprise accounting information in the event of damage. Preparing security recovery plans in advance, including recovery requirements, application priorities, result records, risk assessments, and risk analyses. It is also essential to perform regular backups of the accounting information system operating environment parameters and accounting data, storing backup data in secure locations, and ensuring timely recovery of accounting data in the event of damage or loss through data backup. To ensure the security of accounting information, operations should strictly follow security plans to respond to accounting information system security incidents, record operation results and anomalies, and continuously modify and improve security recovery plans. Lastly, measures should be taken to strengthen controls over accounting personnel, including increasing efforts in professional training for accounting personnel to fundamentally enhance their basic qualities. Providing guidance and supervision to accounting personnel on system operation procedures, ensuring they are familiar with the entire accounting system operation process to reduce accounting risks caused by operational errors. Regular risk prevention education and training for accounting personnel to continually enhance their awareness of system usage security.

Lastly, enhance the system's applicability. With the rapid rise and vigorous development of new generation information technologies such as big data, multidimensional accounting systems are facing increasingly complex data environments. Accounting information systems now extend to all processes of the entire industry chain production and operation, no longer confined to the realm of financial accounting alone. Therefore, it is necessary to strengthen the construction of accounting information systems from both

functional and technical perspectives. This includes improving the management and application capabilities of accounting systems for accounting information in new data environments, as well as the ability to store and process massive data quickly. On one hand, it is imperative to continue strengthening the development and construction of accounting information systems. Encouraging and supporting professional technology companies or industry enterprises to collaborate with technology companies to develop accounting information systems with large data processing and cloud storage functions. By integrating front-end information technology services with accounting information systems, the data management functions of multidimensional accounting information system models are enhanced. On the other hand, multidimensional accounting information systems also need to leverage big data technology to build large accounting information data infrastructure platforms, effectively connecting various information systems within the enterprise to achieve more rapid and efficient real-time, visual data management. This facilitates information resource sharing between departments and systems, effectively addressing the problem of information silos.

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